

Computed Fields

A computed field is a field whose value is derived from a date or number expression, or an interpolated text value. It appears as a standard read-only field but differs in that it is not directly linked to a specific attribute.

Properties

Namespace	Description
Datatype	Number - Display value as number with chosen number format Text* - Displays text with interpolation values.
Expression	Expression to be evaluated.
Format	Applies only to fields with the ' Number ' data type. Format can be of type ' IFS Currency ', ' Decimal ' or ' Percentage '
Label	Label for the field. Field label can be interpolated.
Visible	Determines whether the field is visible. Visible state can be dynamically controlled with a condition.
Column visible	Column visible controls visible state for a table or list column the same way as visible does for other control types.
Column exclude	Excludes the column from a table or list from visible columns. Excluded columns also does not appear in the column chooser.
Searchable	Include the field or column in the search panel

**Fields with the Number data type display values derived exclusively from numerical expressions. If multiple expressions are present, only the value from the first expression is used. For example, in a Number field, `Total Sum: #{Orderlines.sum(Price)}` will display only the sum, omitting the text portion. Any non-numeric content in the expression template is ignored unless the field type is Text.*

Expression

Attributes References

In general, all available data types can be defined by referencing values from another attribute. These attributes may belong to the current record, its parent, or a reference entity. Attribute references can typically be used directly or as parameters within an expression.

Value	Description
MyAttribute	By default, the context is the current record, so an attribute identifier without any namespace refers to the current record (i.e., <code>record.MyAttribute</code>).
parent.SomeAttribute	The <code>parent</code> namespace denotes an attribute on the parent record.
SomeReference.Name	When an entity references other entities, an attribute from the referenced entity can be accessed by prefixing the attribute with the reference name.

Interpolations

An **interpolation value** in a text is a placeholder that is dynamically replaced with actual data at runtime. It allows variables, expressions, or computed values to be inserted into a string, making the text more dynamic. In computed fields, the syntax for interpolation values is divided into `Interpolation Value` and `Number Expression` .

Syntax	Description
<code>\${}</code>	Interpolation value that will insert the value of the referenced attribute into the text at the position.
<code>#{} </code>	Number expressions is used to calculate a number value for a given expression.

Text

Text values can be a combination of static text and interpolation values, with any number of each.

Example: Combine text with a attribute value

```
datatype: Text
expression: Hello, ${Name}
```

Example: Combine text with a calculated value

```
datatype: Text
expression: Total Value: #{Orderlines.sum(Price)}
```

Number Expression

Number expressions supports the usual arithmetic operators `+`, `-`, `*`, `/`, and aggregate functions along with ternary expressions.

Ternary expression

A **ternary expression** (or **ternary operator**) is a concise way to write an **if-else** condition in a single line. It is commonly used to make decisions based on a condition. See [Condition Syntax](#) for more details about condition syntax.

Syntax: `\ ? \ : \`

Example: Calculate price depending if tax should be included or not.

```
datatype: Number
expression: #{IncludeTax ? ( Price * Tax ) : Price}
```

Aggregate functions

A number expression can contain aggregate functions. These are functions that is available on aggregate references.

Function	Description
avg	Calculates average value
count	Counts the number of records
max	Returns the greatest number
min	Returns the smallest number
sum	Calculates the sum

Example: A computed field that shows the total order value can be calculated using the sum function.

```
datatype: Number  
expression: #{Orderlines.sum(Price * Quantity)}  
format: decimal
```

Considerations and Limitations

Using aggregate functions may lead to performance issues since the expression is calculated on the client side, requiring all child records to be loaded.

For performance reasons, aggregate functions are limited to a maximum of 2,000 rows. If the dataset exceeds this limit, the field will display a warning indicating that the value cannot be computed.

The ternary operator can only be used for number expressions. It is not currently possible to use this operator to choose between two different text values. For example, 'Hello #{FullName ? \${FirstName} - \${LastName} : \${FirstName} }' is not supported.